

[5461]-133

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2012 Course) (Semester - I) (402043) (End Sem.)

Time : 2½ Hours]

[Max. Marks : 70

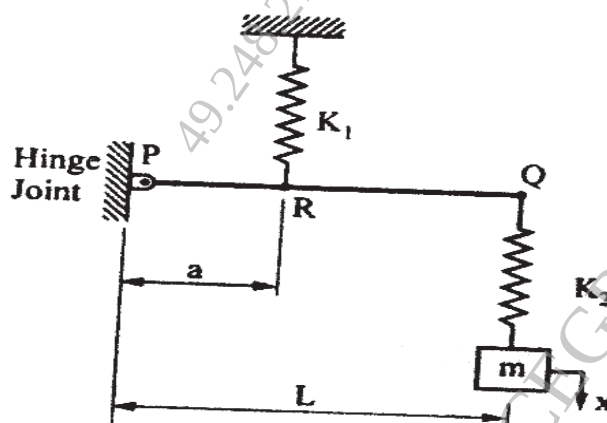
Instructions to the candidates:

- 1) Solve Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8, Q.No.9 or Q.No.10.
- 2) Draw neat diagrams wherever necessary.
- 3) Use of scientific calculator is allowed.
- 4) Assume suitable data, where ever necessary.
- 5) Figures to the right indicate full marks.

Q1) A, B, C & D are four masses attached on a shaft at radii 0.1 m, 0.225 m, 0.15 m and 0.15 m respectively. Planes in which masses revolve are spaced 0.6 m apart and the weights of B, C and D are 10 kg, 5.5 kg and 3.6 kg respectively. Estimate the required mass at A and the relative angular positions of all the four masses so that the shaft is in complete balance. [10]

OR

Q2) a) Determine the natural frequency of the system shown in the fig. Assume the bar PQ to be rigid and weightless. [6]



- b) What is critical speed of shaft? Find out the expression for deflection of the shaft carrying a single rotor without damping. [4]

P.T.O.

Q3) a) A gun barrel, weighing 600 kg has a recoil spring of stiffness 345 N/mm. If the barrel recoils one meter on firing, find

- i) The initial recoil velocity of the gun
- ii) The critical damping coefficient which is engaged at the end of the recoil stroke.

Assume no energy is lost in the recoil of the barrel. [6]

b) Define following terms. [4]

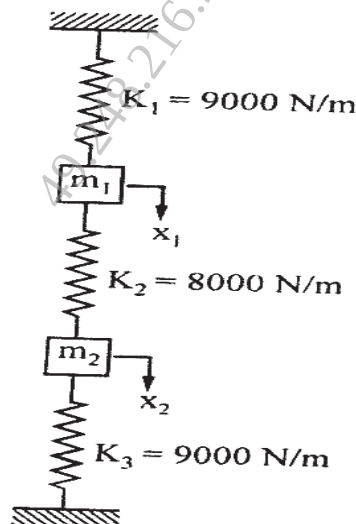
- i) Damping Ratio
- ii) Logarithmic Decrement
- iii) Damping Factor
- iv) Quality Factor

OR

Q4) a) When a single cylinder engine of total mass 300 kg is placed on the four springs, each spring is compressed by 2 mm. A dashpot, offering 400 N of damping force at relative velocity of 200 mm/sec, is attached to the engine to damp out the vibrations. The reciprocating mass of engine is 20 kg and stroke of the piston is 130 mm. If the engine is running at 1500 rpm find the amplitude of steady state vibrations. [6]

b) Explain frequency response curve with neat diagram. [4]

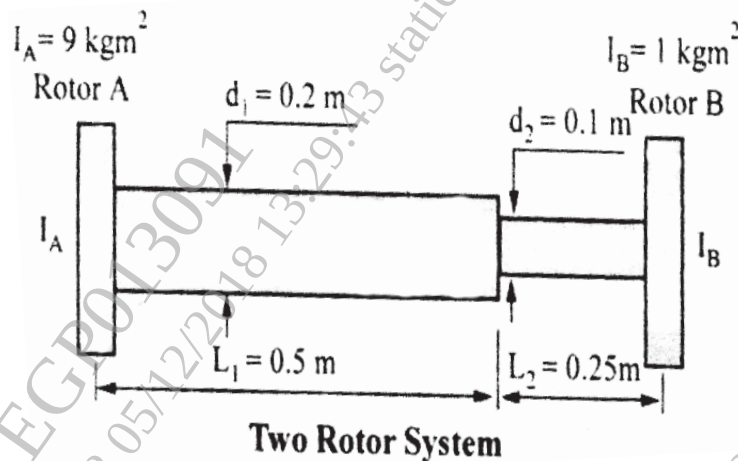
Q5) a) For the system shown in following fig, both the masses are displaced 8 mm, the upper mass in downward direction whereas the lower mass in upward direction. Find the resultant motion of masses. Assume $m_1 = m_2 = 10$ kg. [12]



b) Explain torsionally equivalent shaft. [4]

OR

- Q6) a)** For a 2 rotor system shown in following fig, find the natural frequency and the position of node. Assume $G = 80 \times 10^9 \text{ N/m}^2$. **[12]**



- b)** Explain with neat diagram mathematical model of a quarter car. **[4]**

- Q7) a)** What is vibration isolation? Explain any four isolating materials along with their industrial applications. **[6]**

- b)** Draw and explain transmissibility versus frequency ratio curves. **[6]**

- c)** A vibrometer has a natural frequency of 5 rad/sec and a damping factor of 0.2. The instrument is used to measure a vibrating body having a harmonic frequency of 45 rad/sec. The difference between the maximum and minimum reading is 7 mm. Find the amplitude of motion of vibrating body. **[6]**

OR

- Q8) a)** A device used to measure torsional acceleration consists of a ring having a moment of inertia of 0.05 kgm^2 connected to a shaft by spiral spring having a stiffness of 1 Nm/rad and a viscous damper having a constant of 0.15 Nm-sec/rad . When the shaft vibrates with a frequency of 25 cpm, the relative amplitude between the ring and the shaft is found to be 3° . What is the maximum acceleration of the shaft? **[6]**

- b)** What is vibration absorber? Name different types of absorbers used and explain any one in detail. **[6]**

- c)** What are the different types of vibration pick ups? Explain any one in detail. **[6]**

- Q9) a)** What is a sound field? Differentiate between the free field and diffuse field. [6]
- b) Determine the total sound pressure level due to 90,95,88 dB sound pressure level, arising due to three different sound sources. [4]
- c) What is the decibel scale? Why is a logarithmic scale chosen for sound measurement? [6]

OR

- Q10)a)** An operator in textile mill is operating six machines. The sound pressure levels of the machines at his position are 60 dB, 56 dB, 62 dB, 53 dB, 51 dB and 54 dB respectively. What is the total noise level when all machines are working. [6]
- b) Define following terms: [4]
- i) Acoustic impedance
 - ii) Sound pressure level
 - iii) Sound intensity
 - iv) Sound power level
- c) What is sound enclosure? Describe any one type of sound enclosure.[6]

